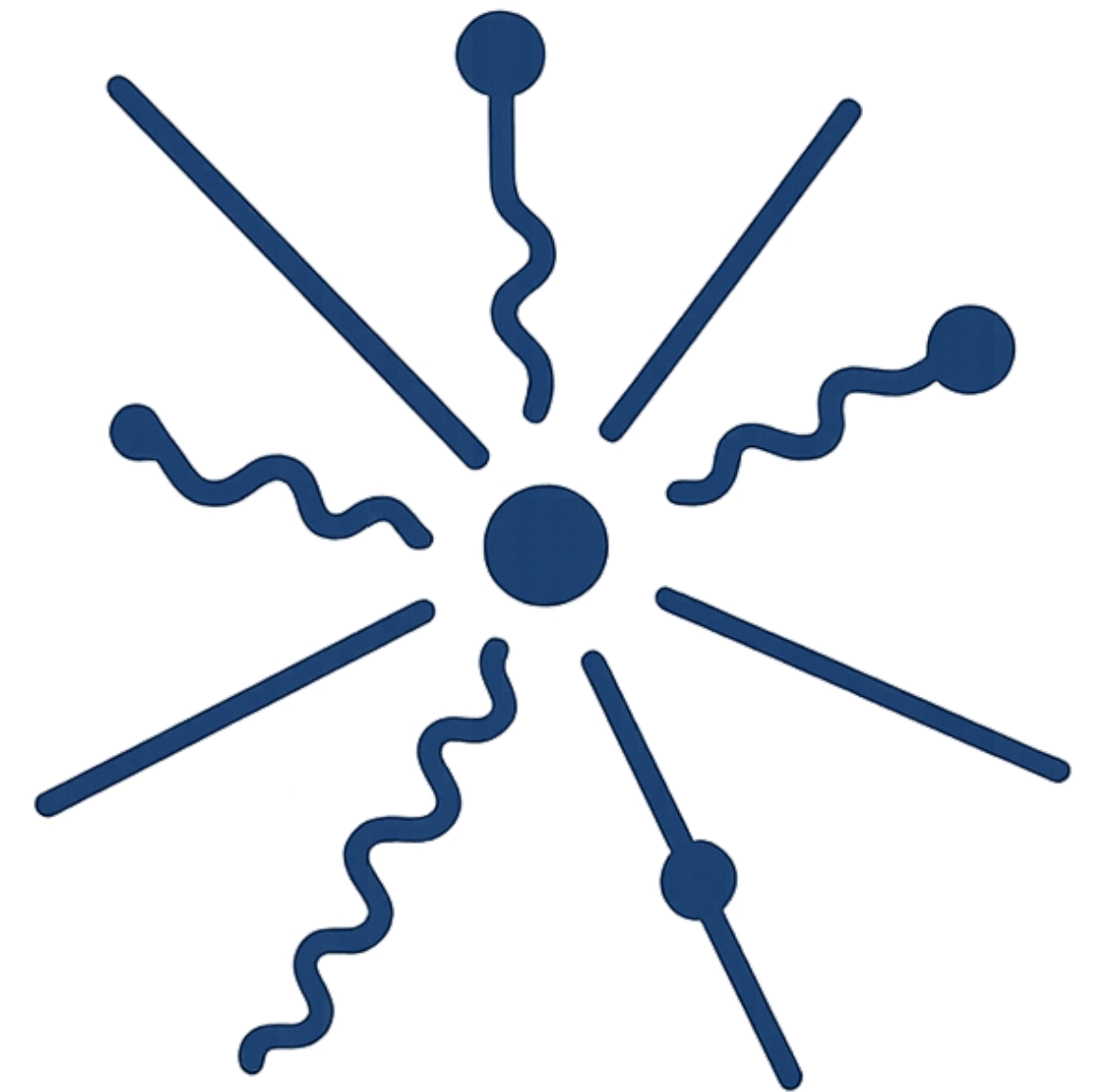
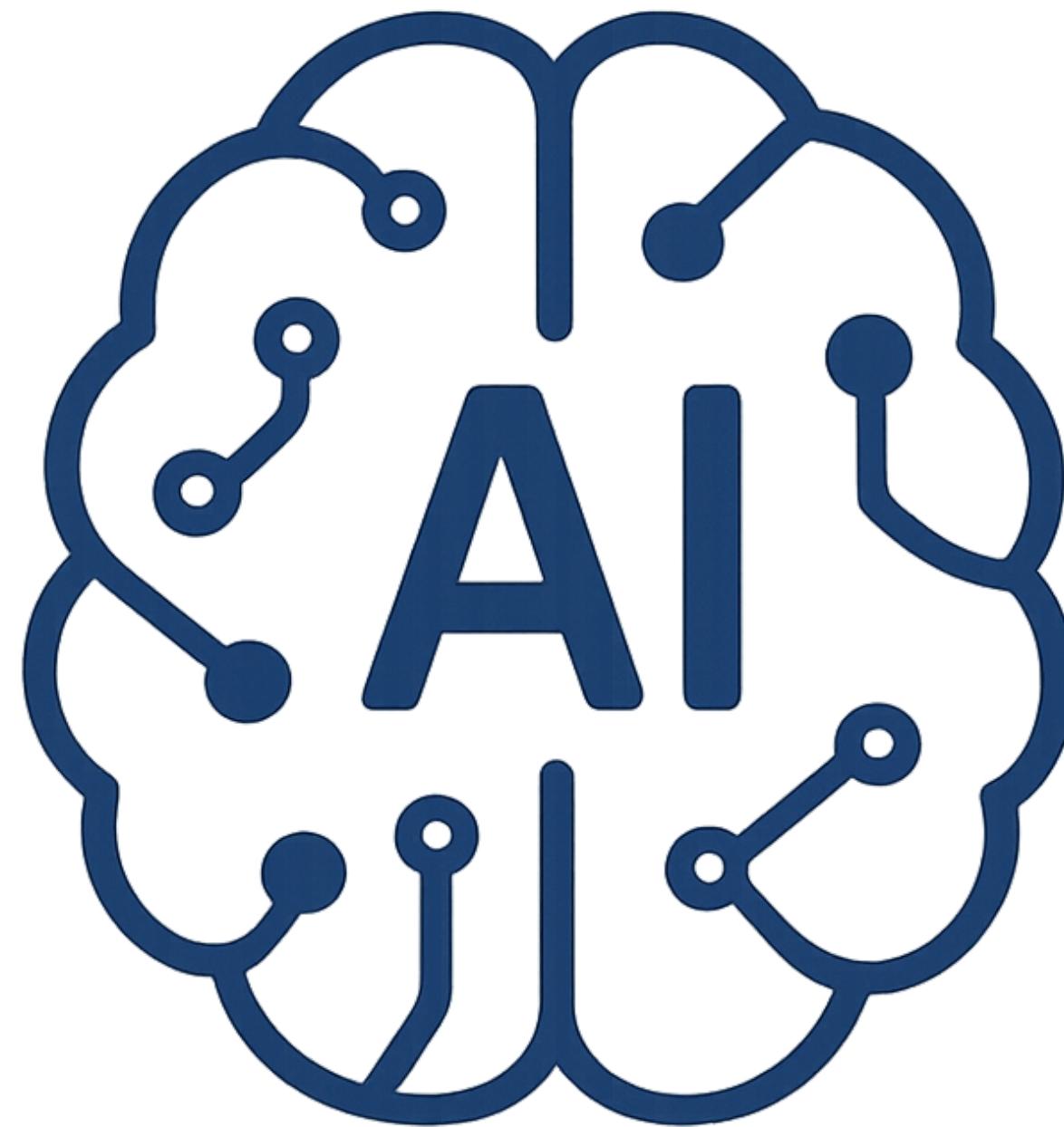
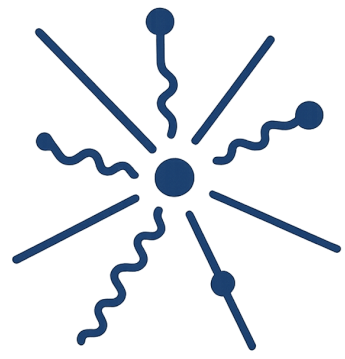
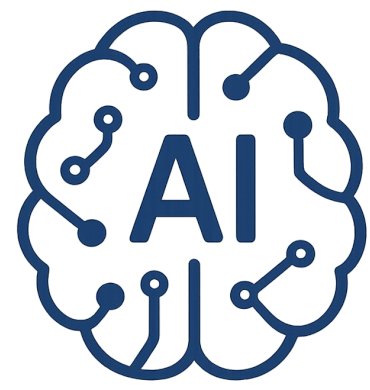


Introduction to

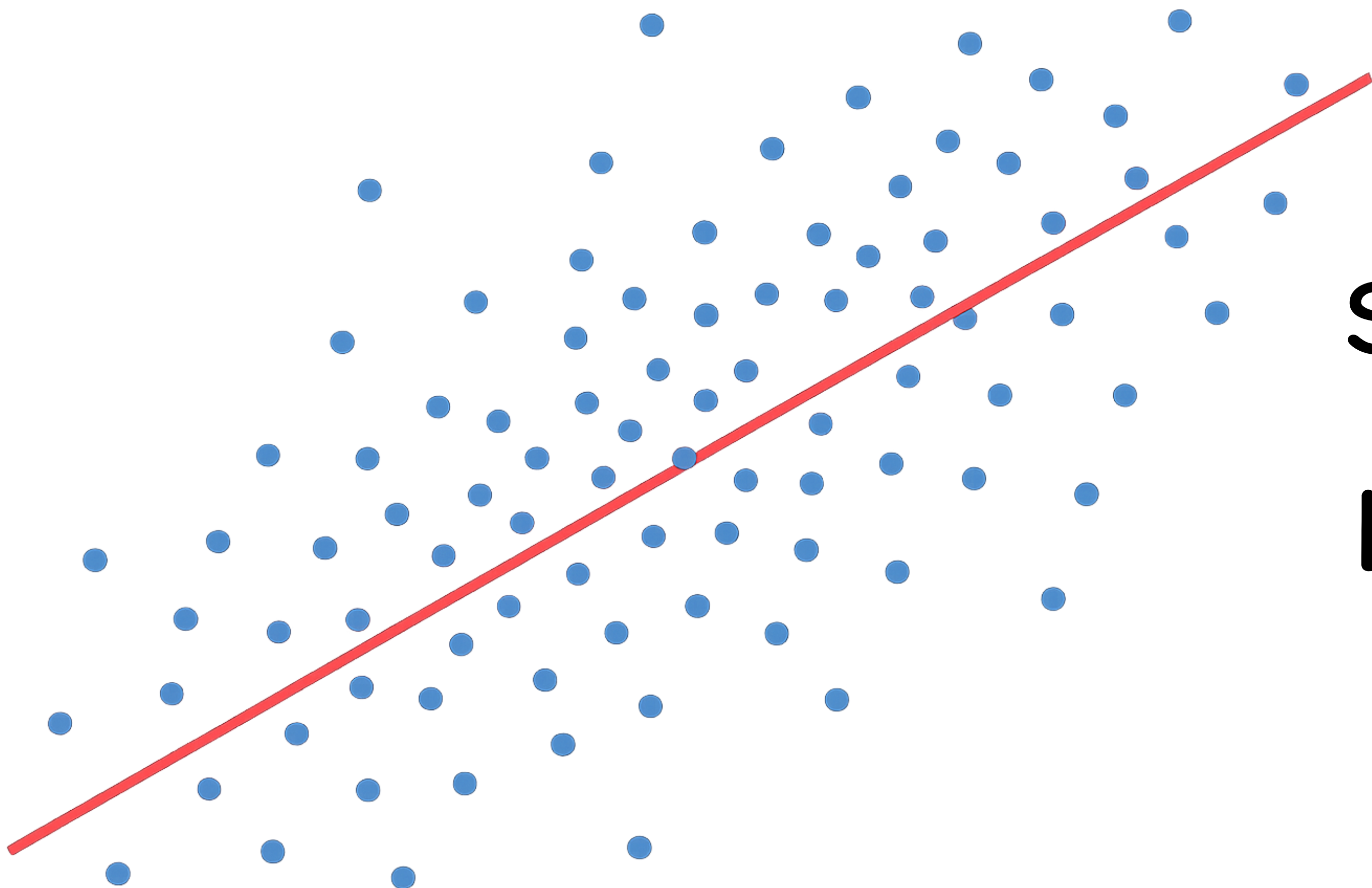
AI-Driven HEP

S. A. Fard - School of Physics (IPM)



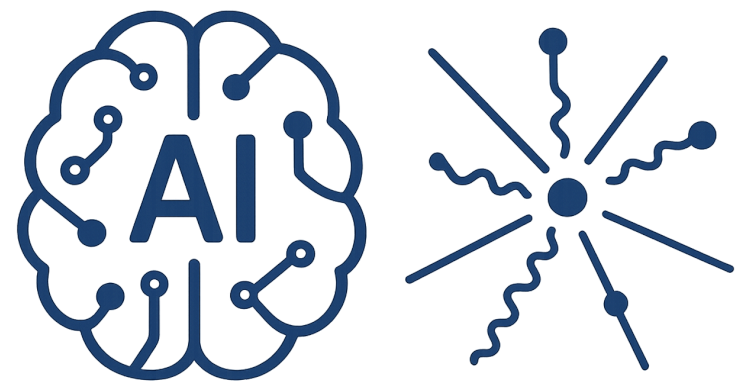


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Session 8

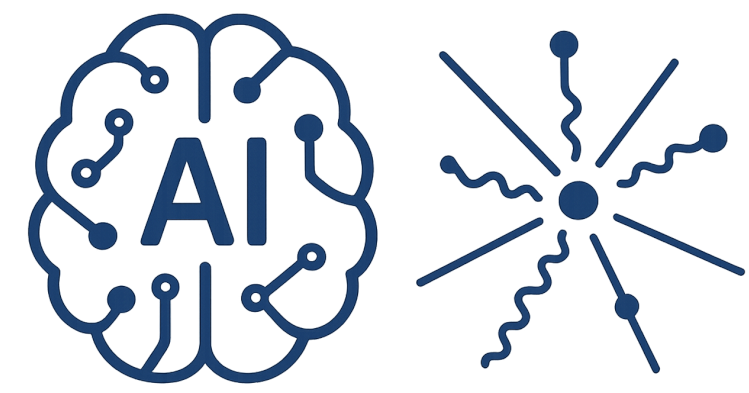
Linear Regression



AI-Driven HEP 8

Reference

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- [2] Hastie, Trevor. "The elements of statistical learning: data mining, inference, and prediction." (2009)
- [3] Bishop, Christopher M., Pattern recognition and machine learning. springer, (2006)
- [4] Chollet, Francois. Deep learning with Python. (2021)
- [5] Shalev-Shwartz, Shai, and Shai Ben-David. Understanding machine learning: From theory to algorithms. (2014)



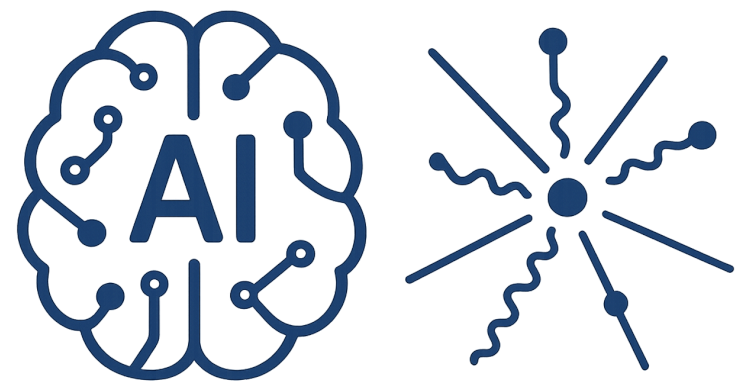
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Linear Regression



Linear Regression

Baseline Model for Comparison

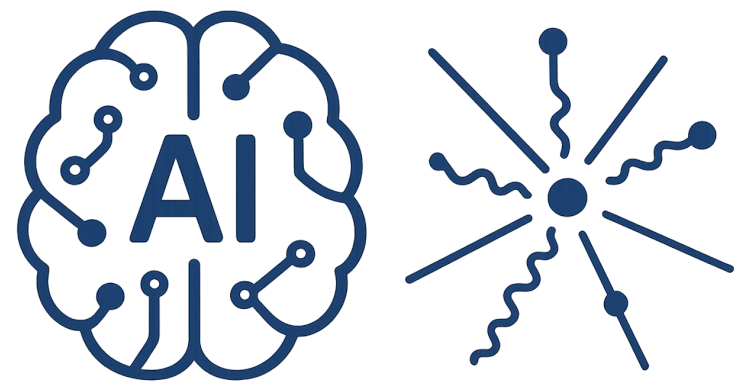


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Linear Regression

Baseline Model for Comparison

Fast and Computationally Efficient



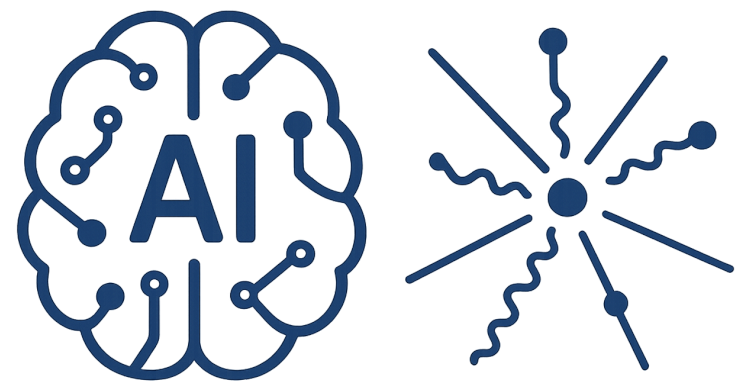
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Linear Regression

Baseline Model for Comparison

Fast and Computationally Efficient

Good Approximation of Local Relationships

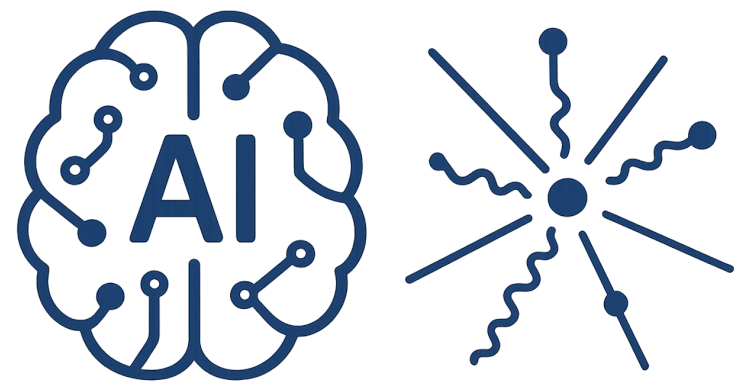


AI-Driven HEP 8

Linear Regression

Simple linear regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$



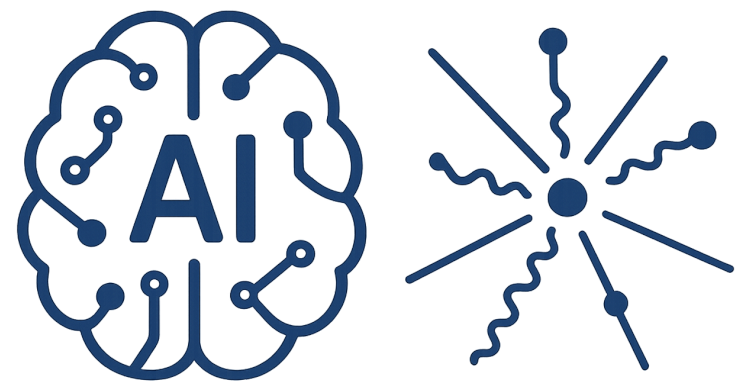
AI-Driven HEP 8

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Estimating the Coefficients



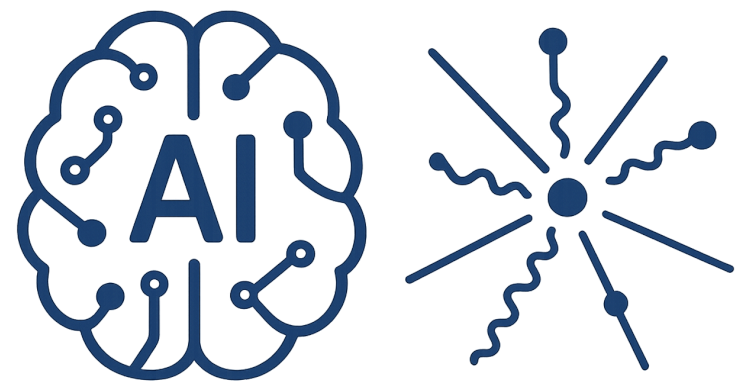
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Linear Regression

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Estimating the Coefficients ?



AI-Driven HEP 8

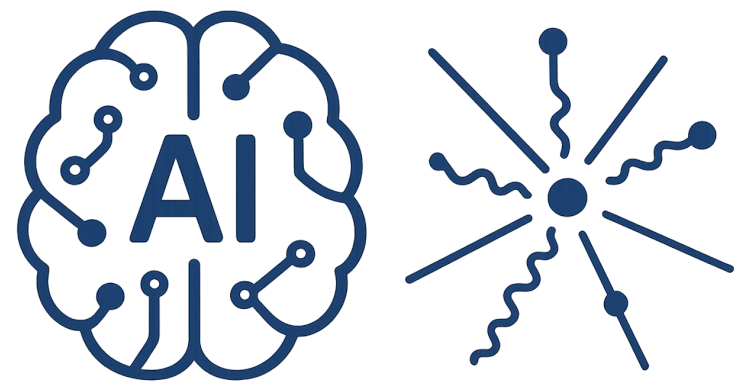
Linear Regression

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Estimating the Coefficients

MLE - LSE



Linear Regression

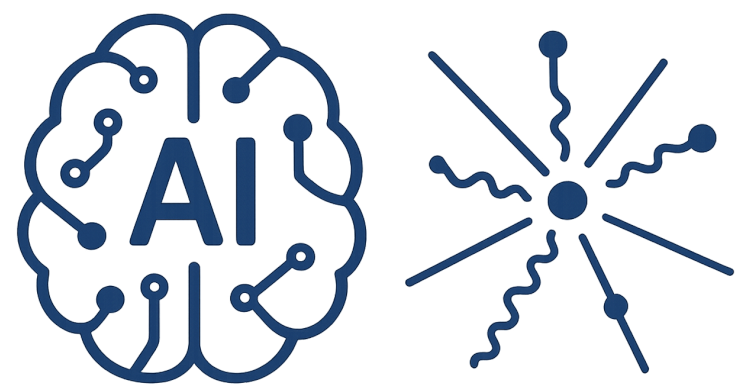
Simple linear regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Estimating the Coefficients

Residual Sum of Squares

$$\text{RSS} = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \cdots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2$$



Linear Regression

Simple linear regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

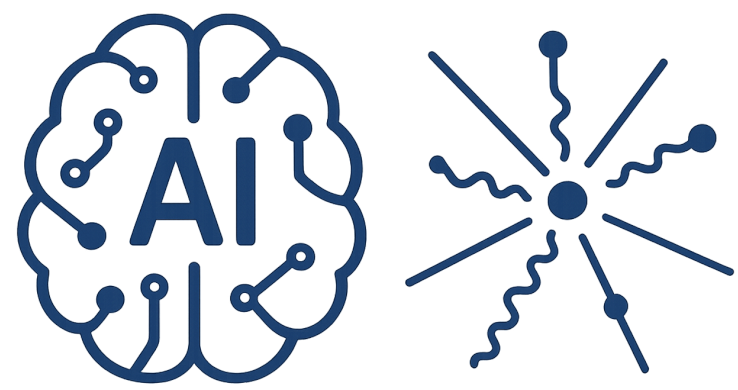
Estimating the Coefficients

Residual Sum of Squares

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$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$



Linear Regression

Simple linear regression

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Estimating the Coefficients

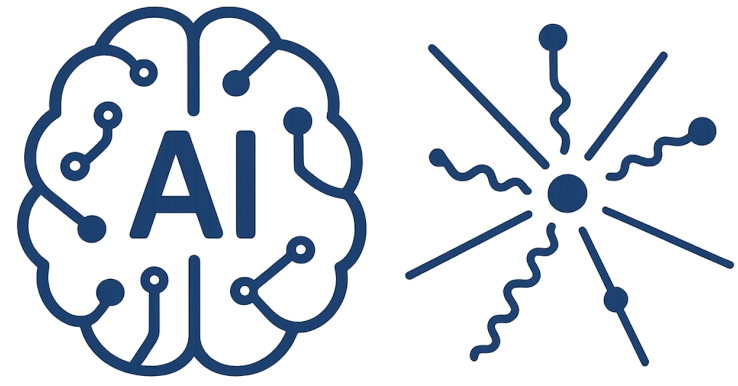
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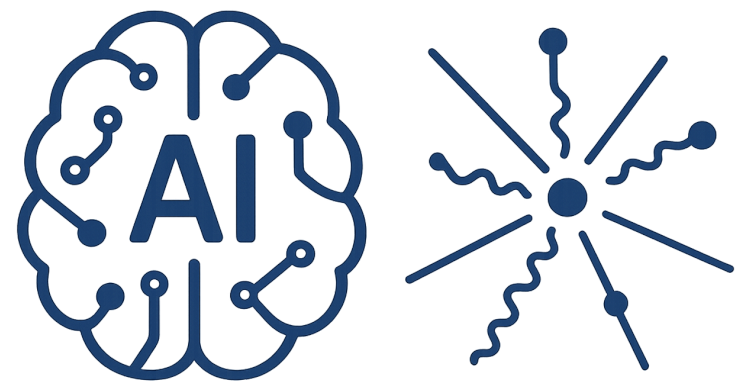
$$S_{yy} = \sum_i (y_i - \bar{y})^2 \quad S_{xx} = \sum_i (x_i - \bar{x})^2$$
$$S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$$



AI-Driven HEP 8

Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_{p-1} X_{p-1} + \epsilon$$



Multiple Linear Regression

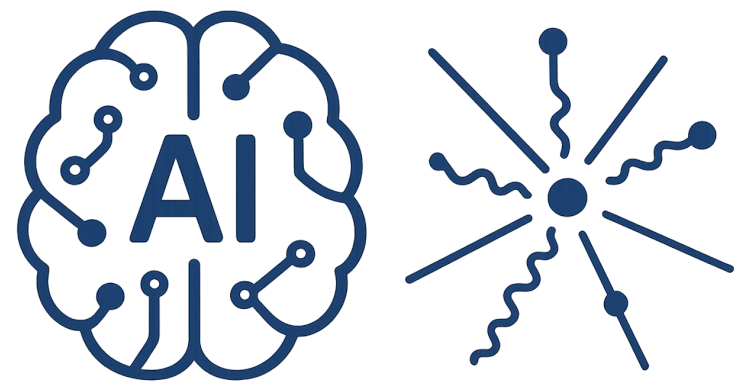
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$$Y = X\beta + \epsilon$$

X : $n \times p$ matrix (one for first column)

β : $p \times 1$ vector of coefficients

ϵ, Y : $n \times 1$ vector of errors



Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_{p-1} X_{p-1} + \epsilon$$

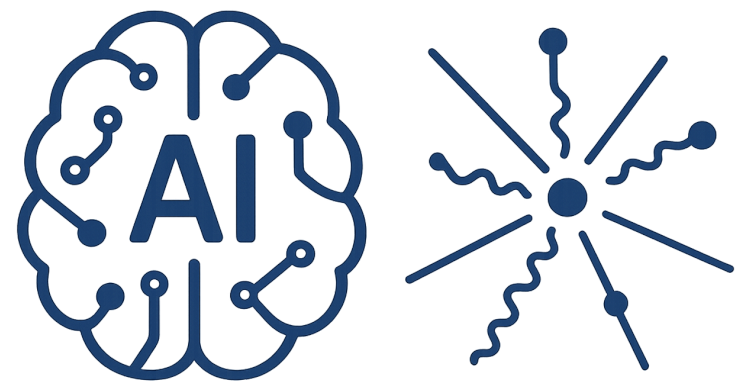
$$Y = X\beta + \epsilon$$

$X : n \times p$ matrix (one for first column)

$$e(\beta) \equiv Y - X\beta$$

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Multiple Linear Regression

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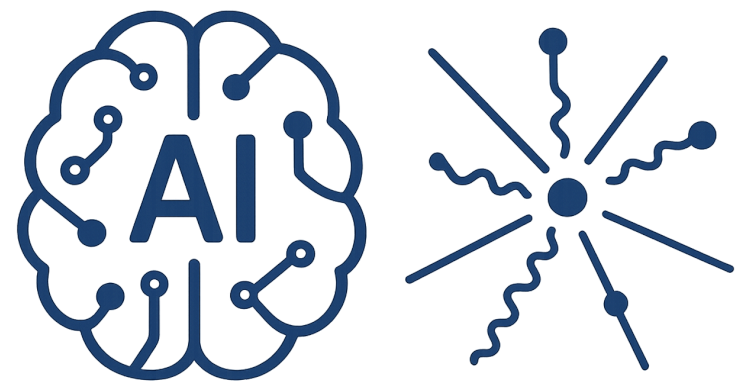
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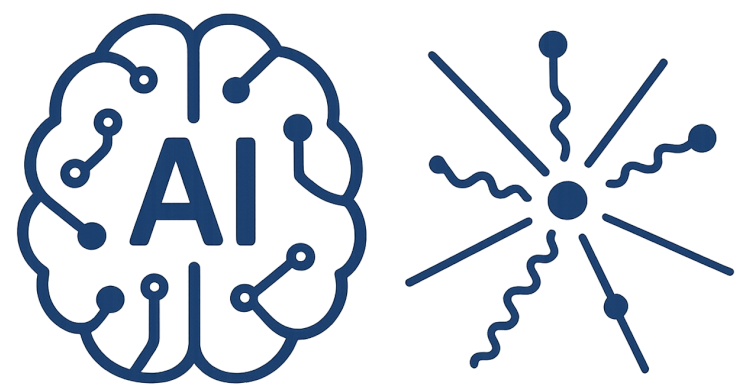
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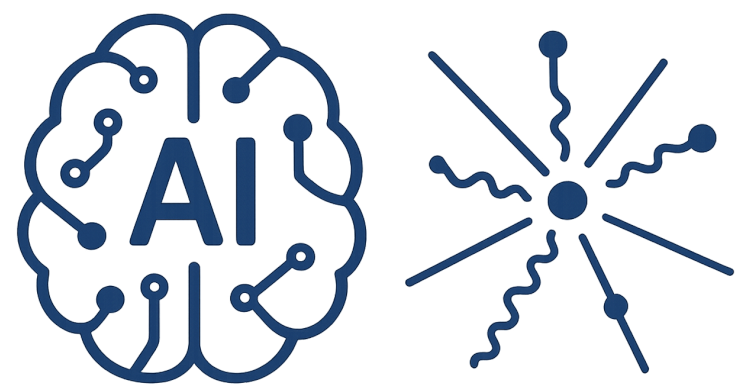
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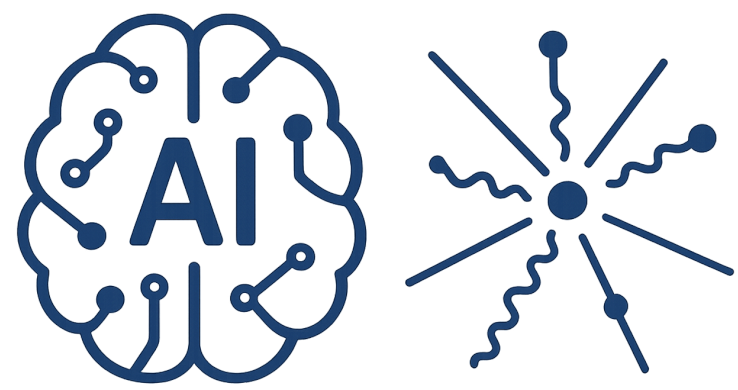
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What about error estimation or what if it is given ?



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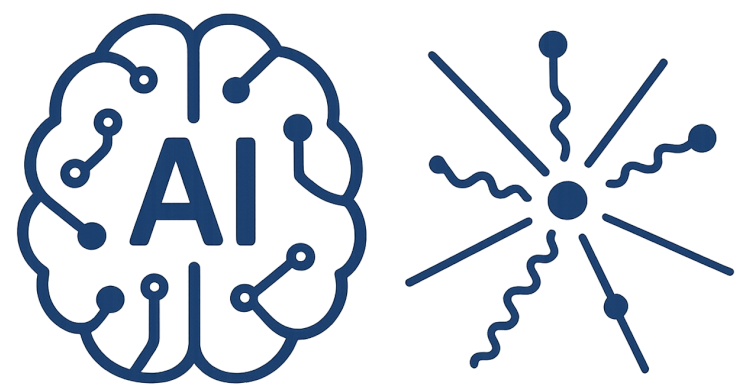
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What about error estimation or what if it is given ?

The nightmare of “model the uncertainty”



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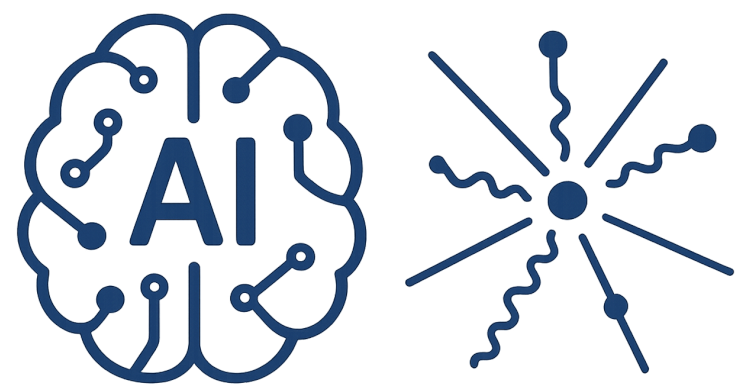
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$$\frac{\partial \text{RSS}}{\partial \beta} = -2X^\top Y + 2X^\top X\beta$$

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$

$$\text{Ex1 : Show that } \hat{\sigma}^2 \equiv \frac{\text{RSS}(\hat{\beta})}{n - p}$$

is an unbiased estimator of σ^2 , provided that ϵ_i are unknown iid $\mathcal{N}(0, \sigma^2)$



Multiple Linear Regression

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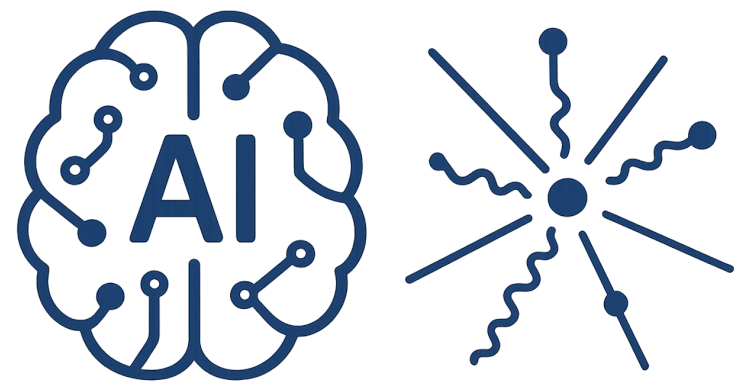
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If σ^2 is given then $\frac{\text{RSS}(\hat{\beta})}{\sigma^2}$ must be $\sim n - p$

Goodness of fit criteria



Multiple Linear Regression

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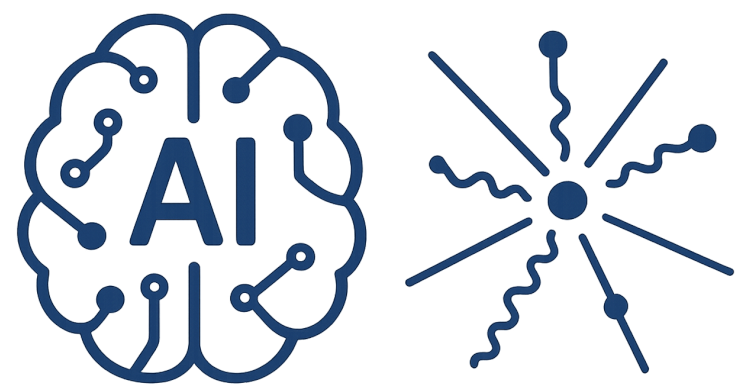
$$\frac{\partial \text{RSS}}{\partial \beta} = -2X^\top Y + 2X^\top X\beta$$

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Ex2 : Starting from

$$\text{RSS}_\Sigma(\beta) \equiv e^\top \Sigma^{-1} e$$

Show that $\hat{\beta} = (X^\top \Sigma^{-1} X)^{-1} X^\top \Sigma^{-1} Y$



Multiple Linear Regression

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Challenges

X^{-1} must be determined numeric

How to generalize to cover more complicated functions